

crowsnest源码安装

1. 背景

1.1 crowsnest是什么？

- Webcam Service for multiple Cams
- A wrapper script for webcam streaming on Raspberry Pi OS Lite images like [MainsailOS](#). Mainly written in bash.
- Mainsail-Crew开发组的摄像头工具，可以将各种摄像头以流的方式通过nginx代理处理，供klipper 3D打印机远程监控使用。

1.2 为什么要源码安装？

因为kiauh自动化安装可能会因为缺少系统依赖而按照失败。

1.3 为什么要本篇文档？

因为源码安装会有很多坑。

2. 安装步骤

2.1 获取crowsnest

代码块

```
1 git clone https://github.com/mainsail-crew/crowsnest.git
```

2.2 安装

2.2.1 Make config

代码块

```
1 cd crowsnest/  
2 sudo make config
```

2.2.2 Make install

代码块

```
1 sudo make install
```

3. 坑

3.1 apt repo配置错误

3.1.1 现象

```
tope@orange4:~/crowsnest$ sudo make install
[sudo] password for tope:
crowsnest - A webcam daemon for multiple cams and stream services.

Ahoy!
Thank you for installing crowsnest ;)
This will take a while ...
Please reboot after the installation has finished.

Running apt-get update first ...

Get:1 https://repo.huaweicloud.com/docker-ce/linux/debian bullseye InRelease [43.0 kB]
Hit:2 https://mirrors.tuna.tsinghua.edu.cn/debian bullseye InRelease
Hit:3 https://mirrors.tuna.tsinghua.edu.cn/debian bullseye-updates InRelease
Get:4 https://repo.huaweicloud.com/docker-ce/linux/debian bullseye/stable arm64 Packages [75.5 kB]
Ign:5 https://mirrors.tuna.tsinghua.edu.cn/debian bullseye-backports InRelease
Err:6 https://mirrors.tuna.tsinghua.edu.cn/debian bullseye-backports Release
      404 Not Found [IP: 101.6.15.130 443]
Get:7 https://security.debian.org/debian-security bullseye-security InRelease [27.2 kB]
Get:8 https://security.debian.org/debian-security bullseye-security/main arm64 Packages [543 kB]
Get:9 https://security.debian.org/debian-security bullseye-security/main armhf Packages [539 kB]
Reading package lists...
E: The repository 'https://mirrors.tuna.tsinghua.edu.cn/debian bullseye-backports Release' does not have a Release file.
Running apt-get update first ... [FAILED]
There is a problem with your apt sources.

Please fix your sytem first and then try again.
make: *** [Makefile:19: install] Error 1
```

3.1.2 原因：repo配置问题或部分不可达

```
tope@orangeipcm4:~/crowsnest$ sudo apt update
Hit:1 https://repo.huaweicloud.com/docker-ce/linux/debian bullseye InRelease
Ign:2 http://deb.debian.org/debian bullseye-backports InRelease
Hit:3 https://mirrors.tuna.tsinghua.edu.cn/debian bullseye InRelease
Err:4 http://deb.debian.org/debian bullseye-backports Release
  404 Not Found [IP: 151.101.90.132 80]
Hit:5 https://mirrors.tuna.tsinghua.edu.cn/debian bullseye-updates InRelease
Hit:6 https://security.debian.org/debian-security bullseye-security InRelease
Reading package lists... Done
E: The repository 'http://deb.debian.org/debian bullseye-backports Release' does
not have a Release file.
N: Updating from such a repository can't be done securely, and is therefore disab
led by default.
N: See apt-secure(8) manpage for repository creation and user configuration detai
ls.
```

3.1.3 解决:

- 使用debian11 bullseye 可用repo模板:
- 注释掉不可用的repo

代码块

```
1  tope@orangeipcm4:~$ cat /etc/apt/sources.list

2  deb https://mirrors.tuna.tsinghua.edu.cn/debian/ bullseye main contrib non-
   free

3  # deb-src https://mirrors.tuna.tsinghua.edu.cn/debian/ bullseye main

4  # contrib non-free

5  deb https://mirrors.tuna.tsinghua.edu.cn/debian/ bullseye-updates main contrib
   non-free

6  # deb-src https://mirrors.tuna.tsinghua.edu.cn/debian/

7  # bullseye-updates main contrib non-free

8  #deb http://deb.debian.org/debian bullseye-backports main contrib non-free

9  # deb-src https://mirrors.tuna.tsinghua.edu.cn/debian/

10 # bullseye-backports main contrib non-free

11 deb https://security.debian.org/debian-security bullseye-security main contrib
   non-free

12 # deb-src https://security.debian.org/debian-security bullseye-security main
   contrib non-free
```

3.2 allow-change-held-packages缺失

3.2.1 现象

```
Checking if device is a Raspberry Pi ... [FAILED]
This device is not a Raspberry Pi therefore camera-streamer cannot be installed .
..
Doing some tests ... [OK]
Installing without camera-streamer ...

Installing dependencies ...

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.9).
findutils is already the newest version (4.8.0-1).
libjpeg-dev is already the newest version (1:2.0.6-4).
pkg-config is already the newest version (0.29.2-1).
git is already the newest version (1:2.30.2-1+deb11u5).
The following additional packages will be installed:
  libcurl4 libevent-core-2.1-7 libevent-extra-2.1-7 libevent-openssl-2.1-7 libevent-
  nt-pthreads-2.1-7 libmd-dev python3-iniparse
The following NEW packages will be installed:
  crudini libbsd-dev libevent-core-2.1-7 libevent-dev libevent-extra-2.1-7 libevent-
  nt-openssl-2.1-7 libevent-pthreads-2.1-7 libmd-dev python3-iniparse
The following held packages will be changed:
  v4l-utils
The following packages will be upgraded:
  bsdtar curl libcurl4 v4l-utils
4 upgraded, 9 newly installed, 0 to remove and 258 not upgraded.
E: Held packages were changed and -y was used without --allow-change-held-packages.
Install dependencies ... [FAILED]
Something went wrong!
Please copy the latest output, head over to

  https://discord.gg/mainsail

and open a ticket in #supportforum...
make: *** [Makefile:19: install] Error 1
```

3.2.2 解决

代码块

```
1 sudo apt upgrade --allow-change-held-packages
2
```

代码块

```
1 sudo apt-mark unhold v4l-utils
```

- 手工安装依赖

代码块

```
1 sudo apt install crudini libbsd-dev libevent-core-2.1-7 libevent-dev libevent-  
extra-2.1-7 libevent-openssl-2.1-7 libevent-pthreads-2.1-7 libmd-dev python3-  
iniparse  
2
```

4. 相关

4.1 crowsnest配置

4.1.1 查看video设备

代码块

```
1 tope@orangebcm4:~$ ll /dev/video*  
2 crw-rw----+ 1 root video 81, 0 Oct 28 11:53 /dev/video0  
3 crw-rw----+ 1 root video 81, 1 Oct 28 11:53 /dev/video1  
4 crw-rw----+ 1 root video 81, 2 Oct 28 11:53 /dev/video2  
5 crw-rw----+ 1 root video 81, 3 Oct 28 11:53 /dev/video3  
6 -rw-rw---- 1 root video 4 Oct 28 09:54 /dev/video-dec0  
7 -rw-rw---- 1 root video 4 Oct 28 09:54 /dev/video-enc0
```

4.1.2 配置cownsnest

```

[crowsnest]
log_path: /home/tope/printer_data/logs/crowsnest.log
log_level: verbose           # Valid Options are quiet/verbose/debug
delete_log: false           # Deletes log on every restart, if set to
    true
no_proxy: false              # If set to true, no reverse proxy is req
    uired. Only change this, if you know what you are doing.

[cam 1]
mode: ustreamer              # ustreamer - Provides MJPG and snapshots
    . (All devices)          # camera-streamer - Provides WebRTC, MJPG
    and snapshots. (only RPiOS + RPi 0/1/2/3/4)
port: 8080                   # HTTP/MJPG stream/snapshot port
device: /dev/video0          # See log for available devices
resolution: 640x480          # <width>x<height> format
max_fps: 30                  # If hardware supports it, it will be for
    ced, otherwise ignored/coerced.
#custom_flags:               # You can run the stream services with cu
    stom flags.
#v4l2ctl:                    # Add v4l2-ctl parameters to setup your c
    amera, see log for your camera capabilities.

[cam 2]
mode: ustreamer              # ustreamer - Provides MJPG and snapshots
    . (All devices)          # camera-streamer - Provides WebRTC, MJPG
    and snapshots. (only RPiOS + RPi 0/1/2/3/4)
port: 8081                   # HTTP/MJPG stream/snapshot port
device: /dev/video2          # See log for available devices
resolution: 640x480          # <width>x<height> format
max_fps: 30                  # If hardware supports it, it will be for
    ced, otherwise ignored/coerced.
#custom_flags:               # You can run the stream services with cu
    stom flags.

```

4.2 摄像头分辨率查看

4.2.1 查看当前视频设备的分辨率和帧率

代码块

```

1  sudo v4l2-ctl --device=/dev/video0 --list-formats-ext
2

```

```

tope@orange4:~$ sudo v4l2-ctl --device=/dev/video2 --list-formats-ext
ioctl: VIDIOC_ENUM_FMT
Type: Video Capture

[0]: 'MJPG' (Motion-JPEG, compressed)
    Size: Discrete 640x480
        Interval: Discrete 0.033s (30.000 fps)
    Size: Discrete 1920x1080
        Interval: Discrete 0.033s (30.000 fps)
    Size: Discrete 1280x720
        Interval: Discrete 0.033s (30.000 fps)
    Size: Discrete 1024x768
        Interval: Discrete 0.033s (30.000 fps)
    Size: Discrete 800x600
        Interval: Discrete 0.033s (30.000 fps)
[1]: 'YUYV' (YUYV 4:2:2)
    Size: Discrete 640x480
        Interval: Discrete 0.033s (30.000 fps)
    Size: Discrete 1920x1080
        Interval: Discrete 0.200s (5.000 fps)
    Size: Discrete 1280x720
        Interval: Discrete 0.100s (10.000 fps)
    Size: Discrete 800x600
        Interval: Discrete 0.050s (20.000 fps)

```

4.2.2 查看设备的详细信息

代码块

```

1  tope@orange4:~$ sudo v4l2-ctl --device=/dev/video0 --info
2  Driver Info:
3      Driver name      : uvcvideo
4      Card type        : USB 2.0 Camera: USB Camera
5      Bus info         : usb-xhci-hcd.7.auto-1.2.4
6      Driver version   : 5.10.160
7      Capabilities     : 0x84a00001
8          Video Capture
9          Metadata Capture
10         Streaming
11         Extended Pix Format
12         Device Capabilities
13     Device Caps      : 0x04200001
14         Video Capture
15         Streaming
16         Extended Pix Format
17 Media Driver Info:
18     Driver name      : uvcvideo
19     Model            : USB 2.0 Camera: USB Camera
20     Serial           : SN0001
21     Bus info         : usb-xhci-hcd.7.auto-1.2.4
22     Media version    : 5.10.160
23     Hardware revision: 0x00000100 (256)

```



```

24         Driver version      : 5.10.160
25 Interface Info:
26         ID                   : 0x03000002
27         Type                  : V4L Video
28 Entity Info:
29         ID                    : 0x00000001 (1)
30         Name                   : USB 2.0 Camera: USB Camera
31         Function               : V4L2 I/O
32         Flags                  : default
33         Pad 0x01000007         : 0: Sink
34         Link 0x02000010: from remote pad 0x1000000a of entity 'Extension 3':
Data, Enabled, Immutable

```

4.2.3 查看摄像头的所有控制选项

代码块

```

1  tope@orangepicm4:~$ sudo v4l2-ctl --device=/dev/video0 --list-ctrls
2          brightness 0x00980900 (int)      : min=-64 max=64 step=1
default=0 value=0
3          contrast 0x00980901 (int)       : min=0 max=64 step=1
default=32 value=32
4          saturation 0x00980902 (int)     : min=0 max=128 step=1
default=64 value=64
5          hue 0x00980903 (int)            : min=-40 max=40 step=1
default=0 value=0
6  white_balance_temperature_auto 0x0098090c (bool) : default=1 value=1
7          gamma 0x00980910 (int)         : min=72 max=500 step=1
default=100 value=100
8          gain 0x00980913 (int)          : min=0 max=100 step=1
default=0 value=0
9          power_line_frequency 0x00980918 (menu) : min=0 max=2 default=1
value=1
10         white_balance_temperature 0x0098091a (int) : min=2800 max=6500 step=1
default=4600 value=4600 flags=inactive
11         sharpness 0x0098091b (int)      : min=0 max=6 step=1
default=3 value=3
12         backlight_compensation 0x0098091c (int) : min=0 max=2 step=1
default=1 value=1
13         exposure_auto 0x009a0901 (menu) : min=0 max=3 default=3
value=3
14         exposure_absolute 0x009a0902 (int) : min=1 max=5000 step=1
default=157 value=157 flags=inactive
15         exposure_auto_priority 0x009a0903 (bool) : default=0 value=1
16  tope@orangepicm4:~$

```